DECODING THE UNSPOKEN LANGUAGE OF HUMAN-AI INTERACTION

[SYSTEM_OVERVIEM]

The Human Communication Eye Protocol (HCEP) v1.2.0

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A Technical Explainer & Vision Architecture

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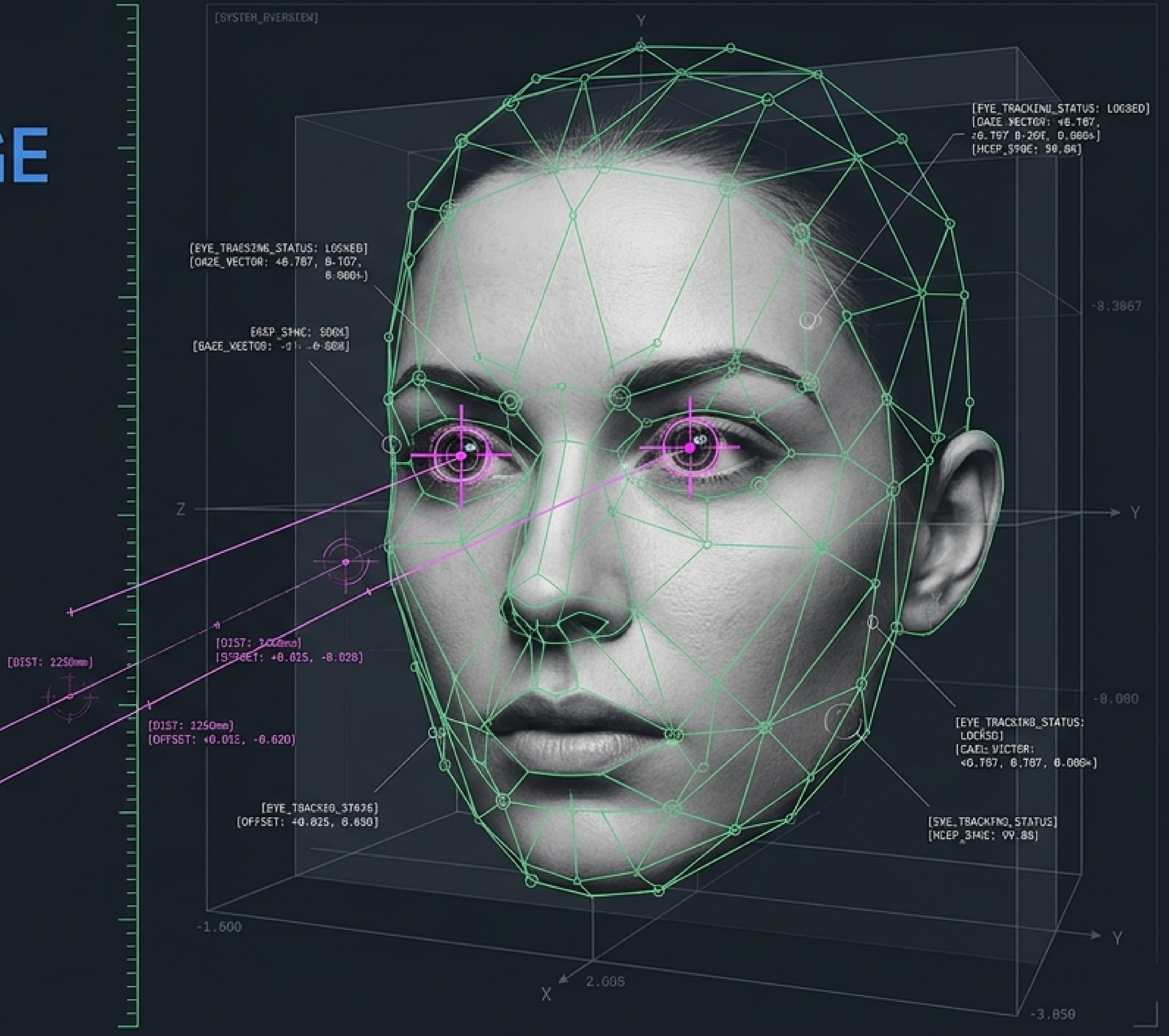
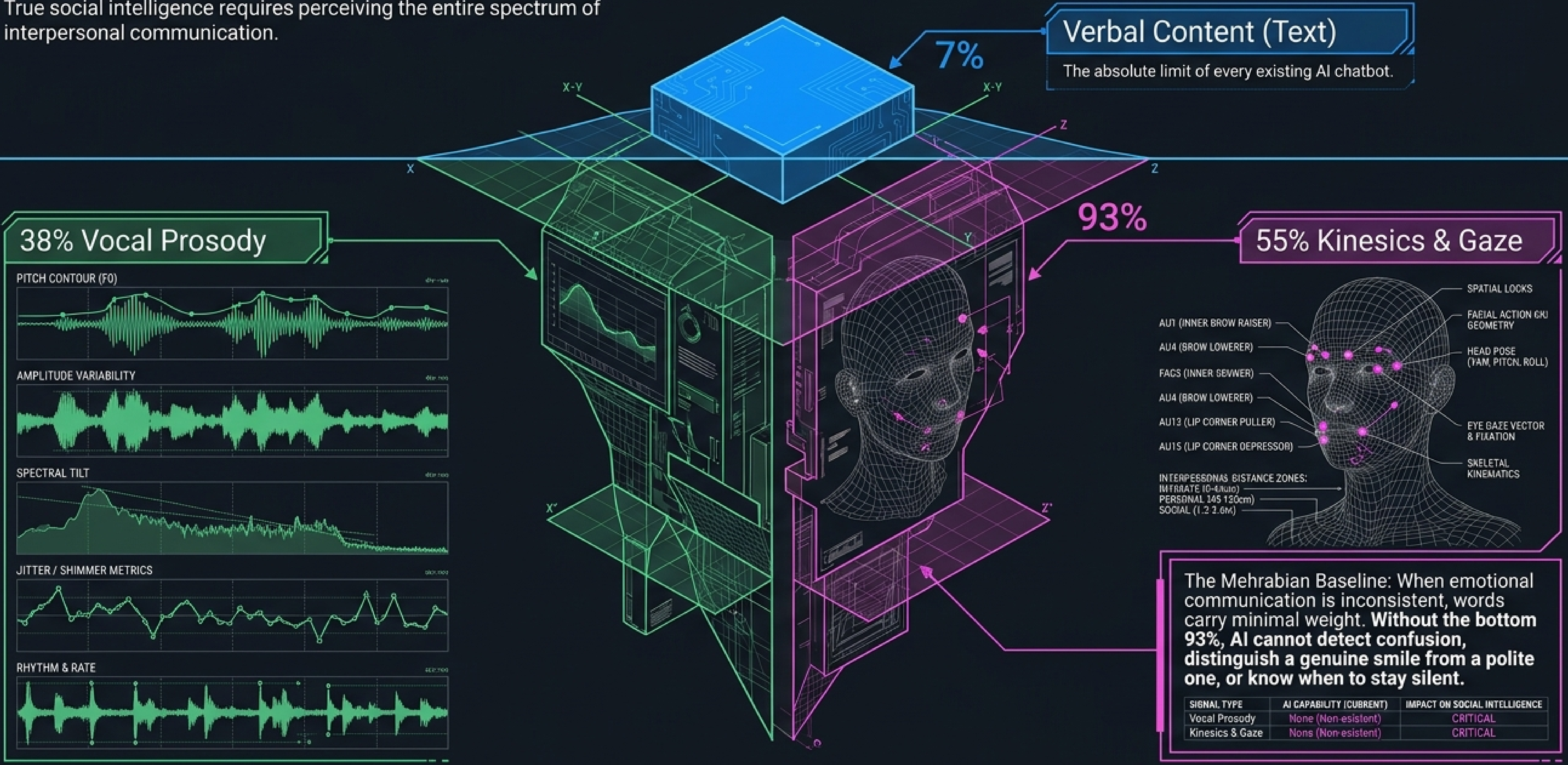


Conversational AI is Stranded at the Tip of the Iceberg

Current LLMs process words—a fraction of actual human meaning. True social intelligence requires perceiving the entire spectrum of interpersonal communication.

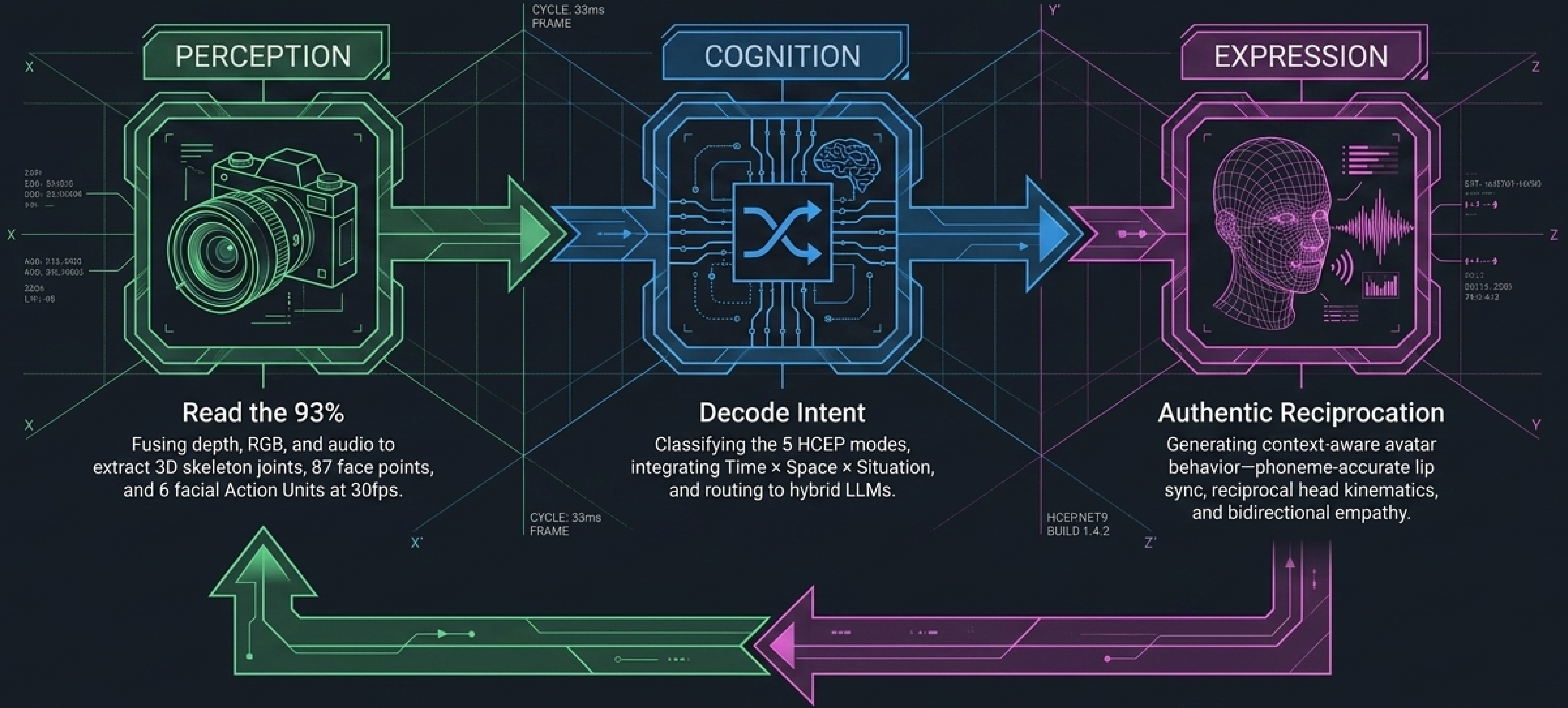


The Mehrabian Baseline: When emotional communication is inconsistent, words carry minimal weight. **Without the bottom 93%, AI cannot detect confusion, distinguish a genuine smile from a polite one, or know when to stay silent.**

SIGNAL TYPE	AI CAPABILITY (CURRENT)	IMPACT ON SOCIAL INTELLIGENCE
Vocal Prosody	None (Non-esistent)	CRITICAL
Kinesics & Gaze	Nons (Non-esistent)	CRITICAL

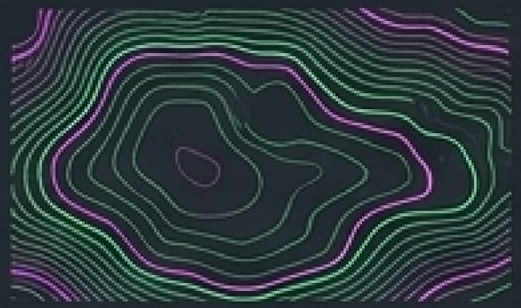
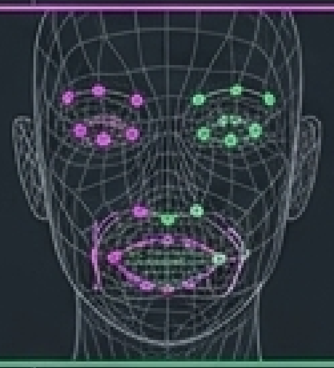
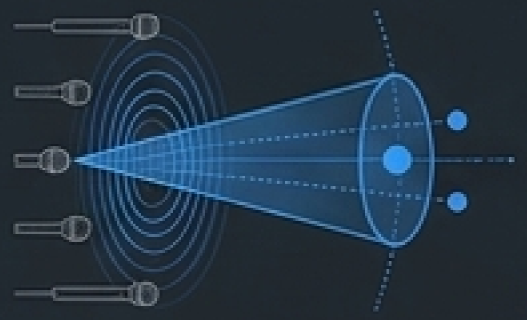
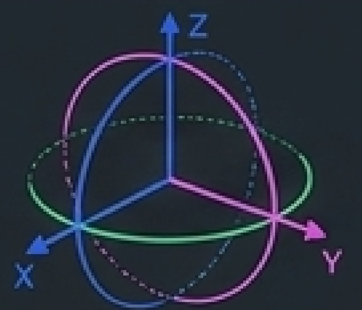
The First Digital Social Nervous System

HCEP (Human Communication Eye Protocol) operationalizes five decades of cognitive neuroscience into a production-ready .NET 9 architecture. It does not just track faces; it completes the full Social Signal Processing loop.



Extracting Behavior from Raw Physics

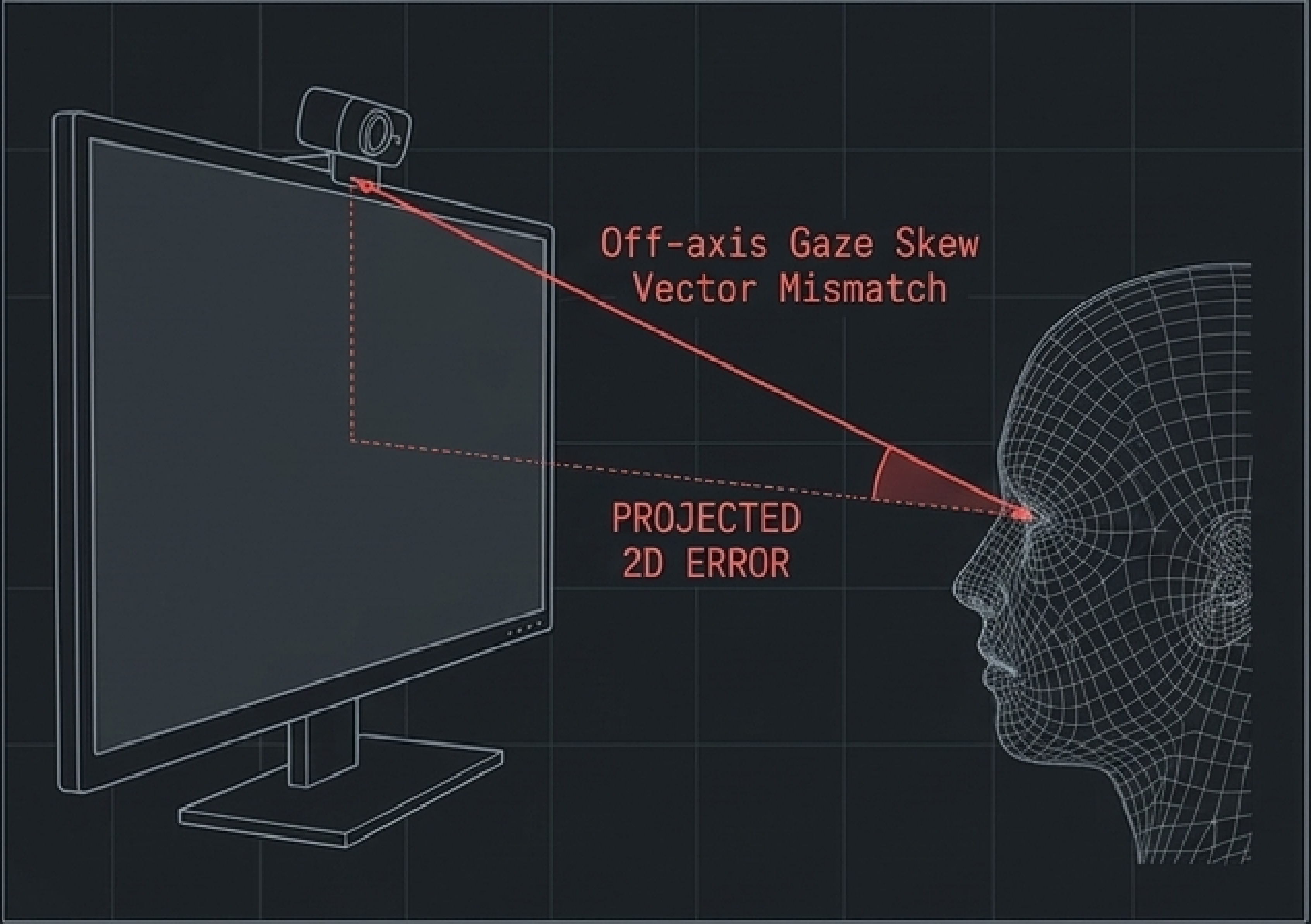
HCEP fuses distinct spatial and optical channels to extract continuous, real-time social biomarkers.

	Raw Data	Extracted Signal
Depth (Kinect/Stereo) 	D13P3 per-pixel depth, 20-joint 3D skeleton.	Proxemic Distance, Postural Approach/Avoidance.
RGB Vision 	BGRA32 at 30fps, 87-point mesh.	6 Action Units (AUs). Discriminates the Duchenne Marker (genuine smile: AU12+AU6) from social masking.
Spatial Audio 	4-mic linear array, 16kHz beamforming.	Voice Activity Detection (VAD), Turn-taking boundaries.
Head Kinematics 	Euler angles (±2-3° precision).	Backchannel Nods, Epistemic Head Shakes, Empathic Tilts.

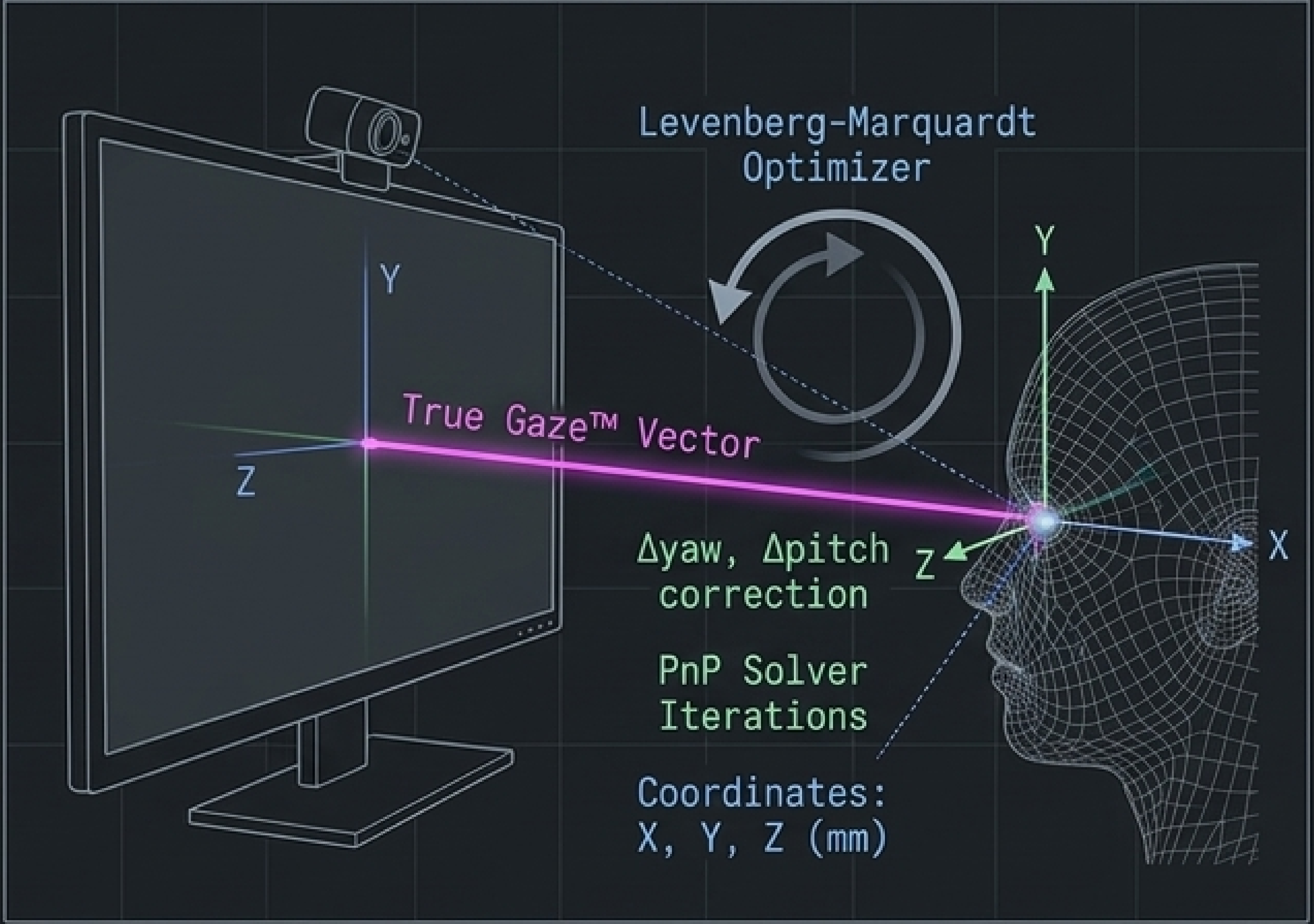
World Space Lock & True Gaze™ Calibration

Off-axis sensors historically cause severe gaze skew. HCEP anchors its coordinate system to physical reality, achieving sub-millimeter stability via an iterative PnP (Perspective-n-Point) solver.

The Flaw: Miscalculated 2D Projection



The HCEP Solution: 3D Intersecting Coordinate Planes



Fuses 6 canonical landmarks to extract 3-stage gaze:
Head Pose -> Eye-in-Head Rotation -> Hybrid Fusion (Exponential Moving Average smoothed)

The 5-Mode Classification Matrix

Beyond binary looking or not looking, HCEP categorizes eye contact into 5 distinct cognitive-emotional states based on temporal dynamics, facial Action Units, and spatial targeting.

Mode	Pattern	AUs/State	AI Strategy
LOGIC	▪ On-face, structured scanning.	▪ AU4 mild / Analytical processing.	▪ Precise, factual, numbered lists.
AFFECT	▪ Social Triangle (eyes <-> mouth).	▪ AU12 + AU6 / Emotional engagement.	▪ Warm, empathetic, feeling-first.
SPIRIT	▪ Sustained mutual gaze (>3s).	▪ Low AU, relaxed / Deep authentic rapport.	▪ Personal, genuine, unstructured.
HEART	▪ Lower-face focus.	▪ AU1+AU4, AU15 / Empathic resonance.	▪ Supportive, validating, caring.
THINK	▪ Gaze aversion (>15° defocus).	▪ Moderate AU4 / Internal processing.	▪ Brief, non-intrusive, space-giving.

Empirical Validation: Performing at the Human Baseline

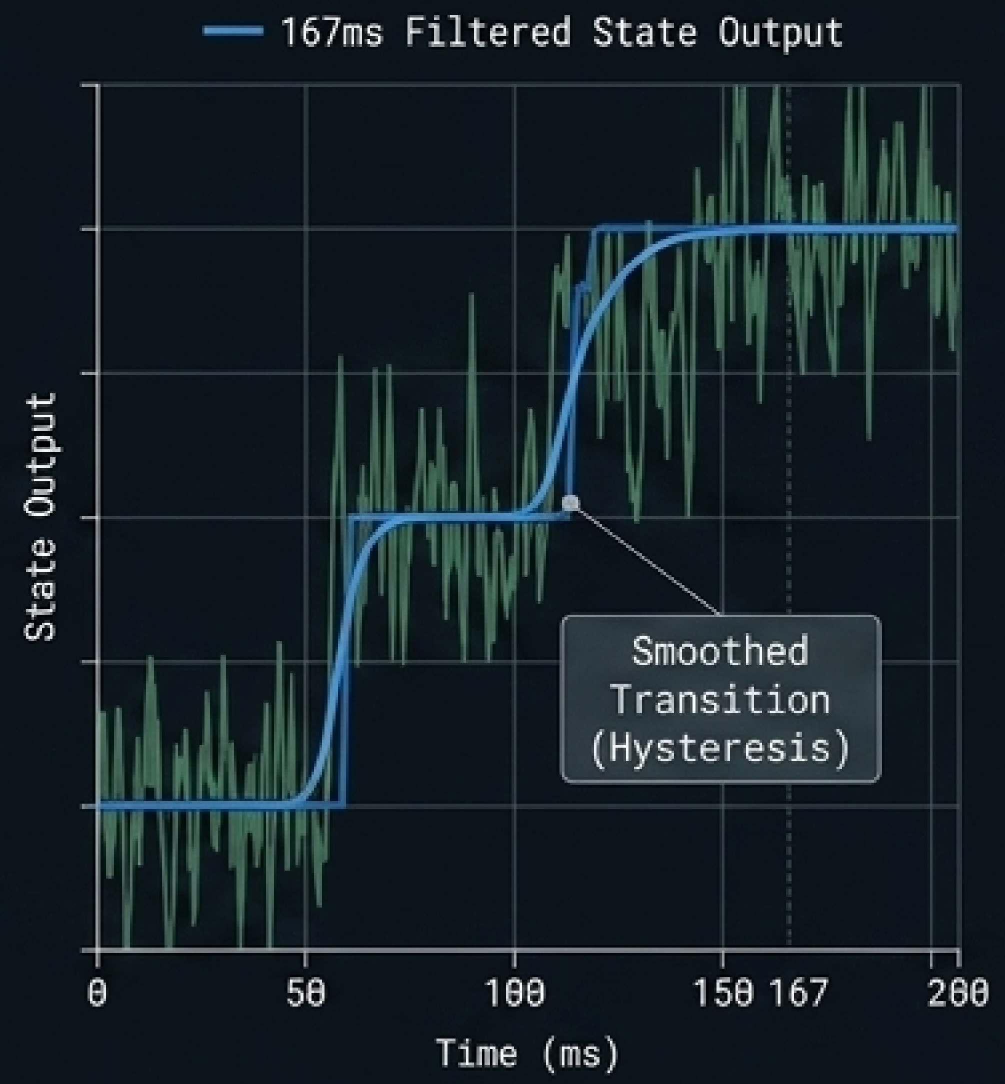
Tested against 6,000 frames of conversational data coded by three independent human annotators, HCEP precisely matches human reliability in deciphering nonverbal intent.

Cohen's Kappa: 0.8084. Scores 0.81-1.00 indicate Almost Perfect Agreement. HCEP reads intent as accurately as human experts agree with each other.

84.55%

Classification Accuracy: 84.55%. Exceeds the 80.0% target baseline.

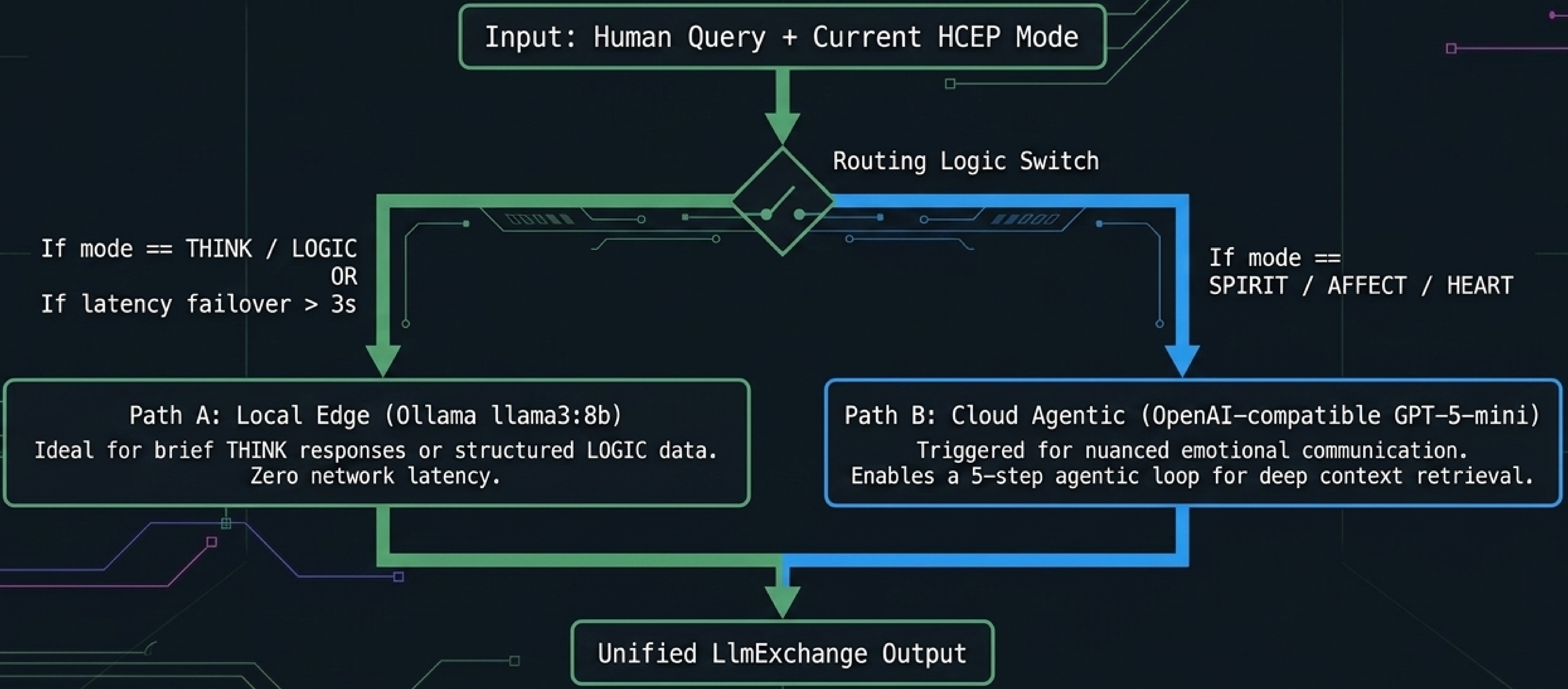
Temporal Stability: 5-frame Hysteresis Smoothing



Demonstrating sub-200ms latency with high stability.

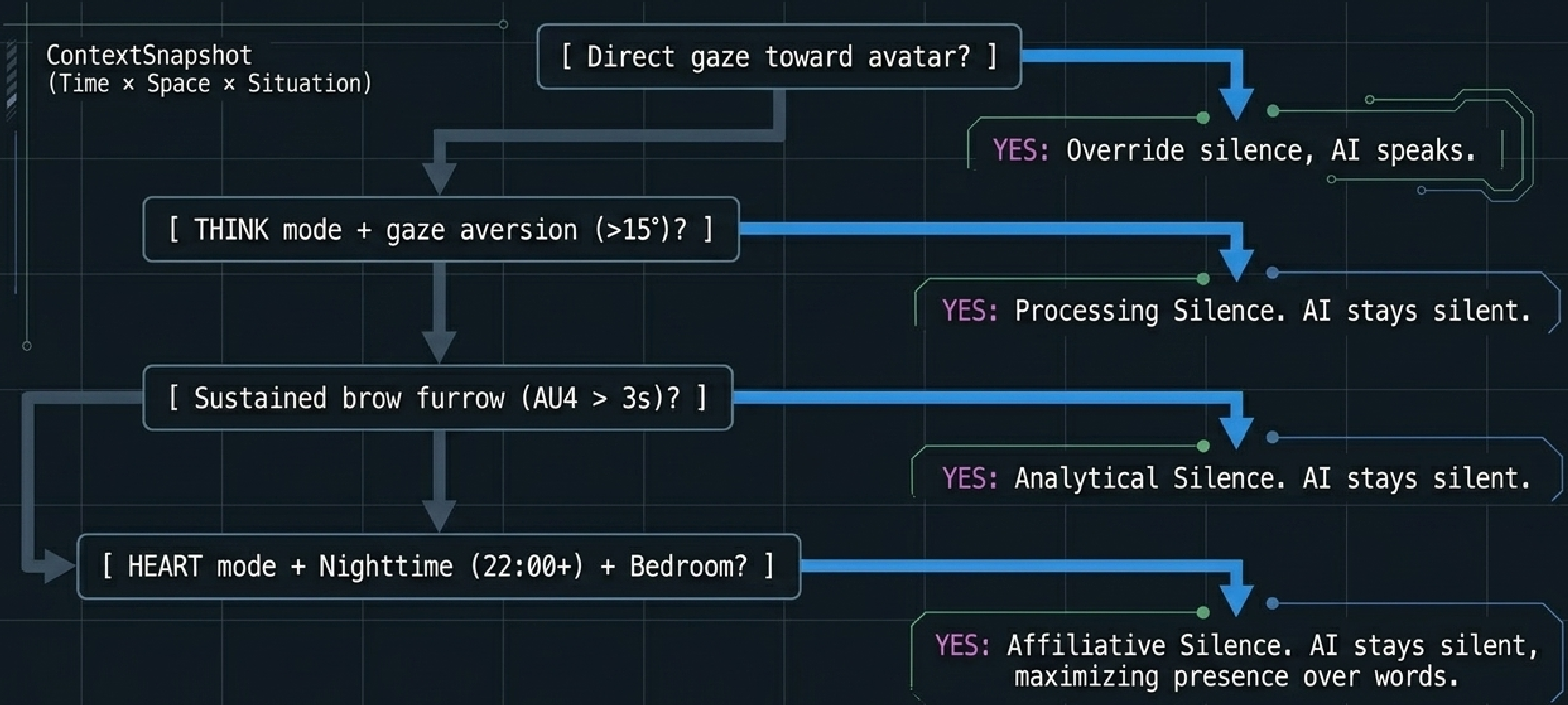
Contextual Routing: The Hybrid LLM Engine

HCEP does not use a single AI brain. The HcepPromptBridge dynamically routes queries between edge and cloud models based on the detected cognitive mode and query complexity.



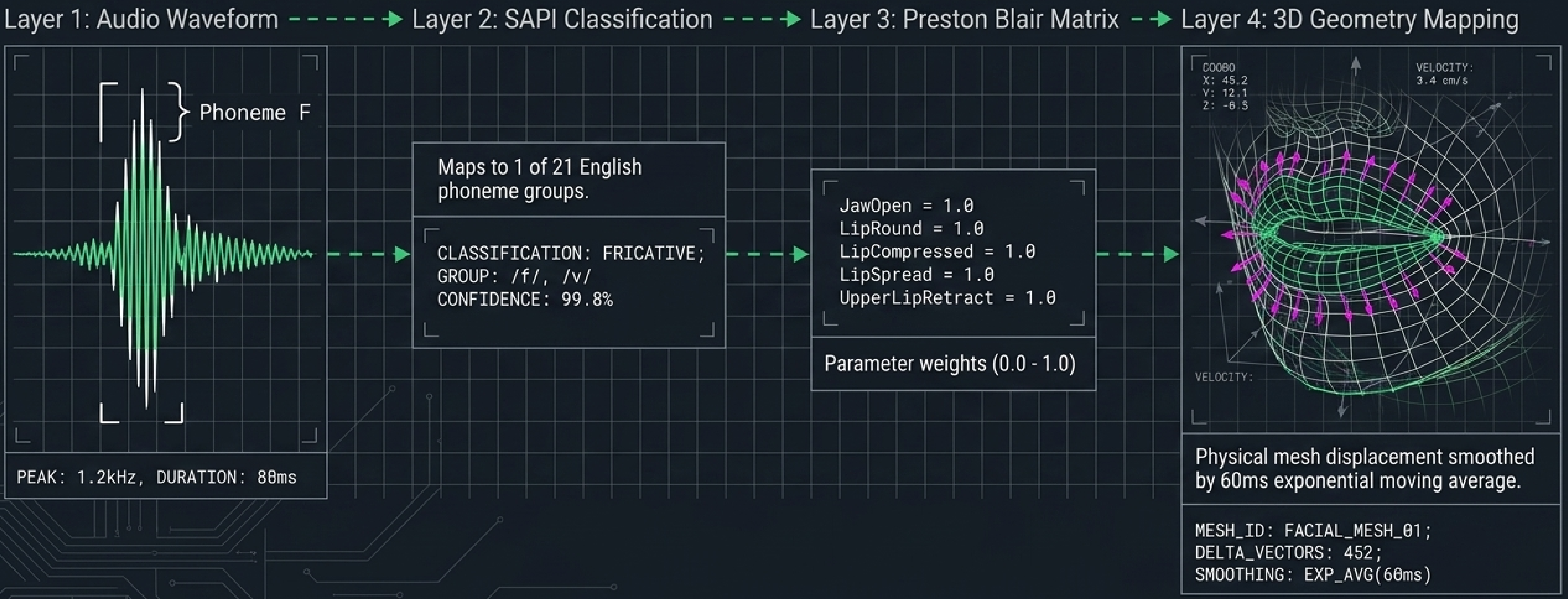
The Silence Protocol: Knowing When Not to Speak

The most sophisticated communicative act is often silence. The SilenceProtocolEvaluator uses 7 priority rules to suppress AI speech during human cognitive processing or empathic vulnerability.



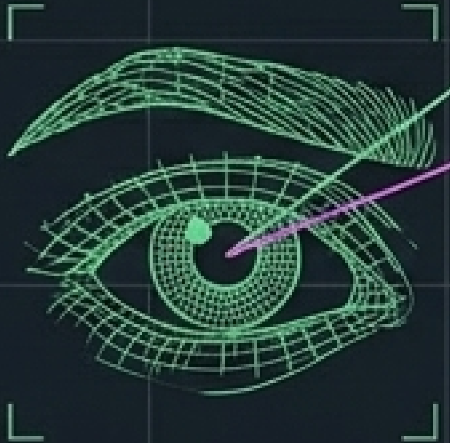
Defeating the McGurk Effect: Phoneme-to-Viseme Anatomy

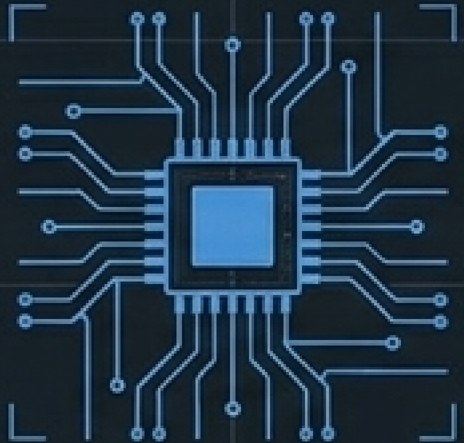
Visual lip movement is processed by the human brain as a first-class speech channel. Inaccurate lip sync actively degrades audio intelligibility and triggers uncanny valley revulsion.

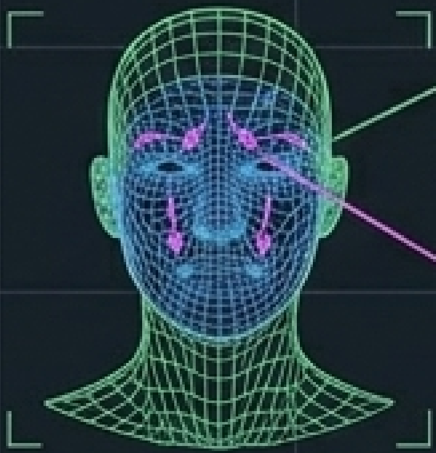


The Full Reciprocation Loop in Practice

How the Digital Social Nervous System processes a single, silent moment of human vulnerability.







Sensor Green

T=0ms (Perception)

User's gaze drops and defocuses. FaceTrackLib detects AU4 (brow lower) + AU15 (lip depressor). HCEP classifies THINK mode transitioning to HEART mode.

T=150ms (Cognition)

SilenceProtocolEvaluator intercepts the state change. It recognizes the user is processing deep emotion. AI output is intentionally suppressed.

T=300ms (Expression)

ExpressionMirror triggers the avatar response. Avatar initiates a gentle forward head tilt, inner brow raise (AU1), and sustained empathic body language.

Impact: The human experiences being seen and given space, without a single word being spoken.

A Production-Ready .NET 9 Architecture

A strict, bottom-up dependency stack utilizing System.Threading.Channels for backpressure-aware, multi-threaded 30fps execution.

APP & UI LAYER	HCEP.App (WPF Dashboard, MVVM, Avatar Rendering, API Host)
	Bounded Channels (DropOldest)
COGNITION LAYER	HCEP.Intelligence (Hybrid LLMs, Agentic Tools) + HCEP.Knowledge (Strategy D UKS)
	Bounded Channels (DropOldest)
PERCEPTION LAYER	HCEP.Vision (ArcFace, Mode Analyzer) + HCEP.Audio (Whisper) + HCEP.Spatial
	Bounded Channels (DropOldest)
SENSOR LAYER	HCEP.Kinect (COM Interop) / Webcams / Simulators
	Bounded Channels (DropOldest)
FOUNDATION LAYER	HCEP.Core (Interfaces, Enums, Models) + HCEP.Telemetry

Immutable Safety & Biometric Compliance

Enterprise-grade deployments require absolute data sovereignty and rigid ethical constraints. HCEP is fortified by design.

Cryptographic Directives

Permanent Active Directives (10 Laws). Validated on startup via SHA-256 hash. If modified, the system locks into a deep diagnostic state to prevent prompt injection.

Enterprise Vaulting

API keys secured via DPAPI and Windows Credential Manager.

Keys exist only in the WCM vault, never in process memory dumps.

Biometric Sovereignty

Full BIPA/GDPR compliance.

Identity extraction requires explicit UI consent, with automated TTL data purge routines.

The HCEP SDK & Interoperability Ecosystem

Deploy real-time social intelligence into robotics, spatial computing, and existing autonomous agents.

Model Context Protocol (MCP)

Anthropic MCP-compliant POST /mcp endpoint for dynamic agentic tool discovery.

Spatial Computing

Native Unreal Engine C++ Actor Components and Unity C# Controllers for real-time MetaHuman avatar gazing.

HCEP Runtime Hub

Microsoft Stack

Native Semantic Kernel plugins mapping HCEP tools directly to enterprise C# agents.

Python AI Frameworks

Drop-in LangChain & LlamaIndex wrappers for injecting real-time human gaze states into LLM context windows.

The End of Dead Eyes: Achieving Social Cognitive Fidelity

Bailenson (2001) proved that social presence in virtual spaces is determined not by graphical realism, but by the behavioral fidelity of gaze and expression.

The Status Quo

Graphical Fidelity.
Beautiful, but functionally dead.

The HCEP Future

Social Cognitive Fidelity.
Truly present, truly aware.

HCEP transforms AI from a text processor into a spatially and socially aware partner. Machines finally speak our unspoken language.